

# Strategic Data Analytics: Tokopaedi Case Study

Remaking Technical Insights into Business Intelligence

---

Presented by Kadek Ngurah Silayoga

---

# **Executive Agenda & Methodology**

Comprehensive evaluation of e-commerce performance across sales, categories, and funnel efficiency.

# Data Ecosystem & Relational Schema

## INTEGRATED DATA MODEL

### The Data Foundation

Tokopaedi's analytical engine is built on six core relational tables: **Order**, **Product**, **Customer**, **Payment**, **Funnel**, and **Transaction details**. This structure allows for multi-dimensional join operations to track the user journey from click to cash.



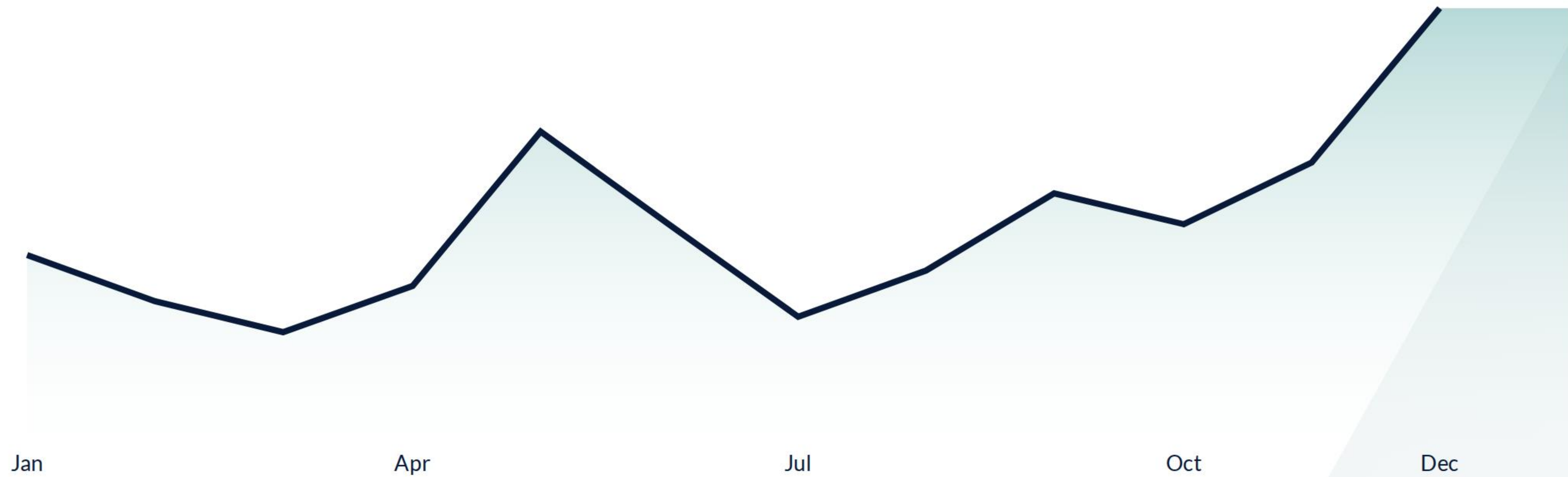
## ANALYTIC OBJECTIVES

### Key Research Areas

- **Growth:** Monthly Revenue Trends (2024).
- **Product:** Volume Analysis by Category.
- **Channel:** Efficiency of Web vs App vs Offline.
- **Funnel:** Organic Conversion Rate metrics.
- **Onboarding:** Time-to-First-Order speed.

# Peak Demand Occurs in Q4 December

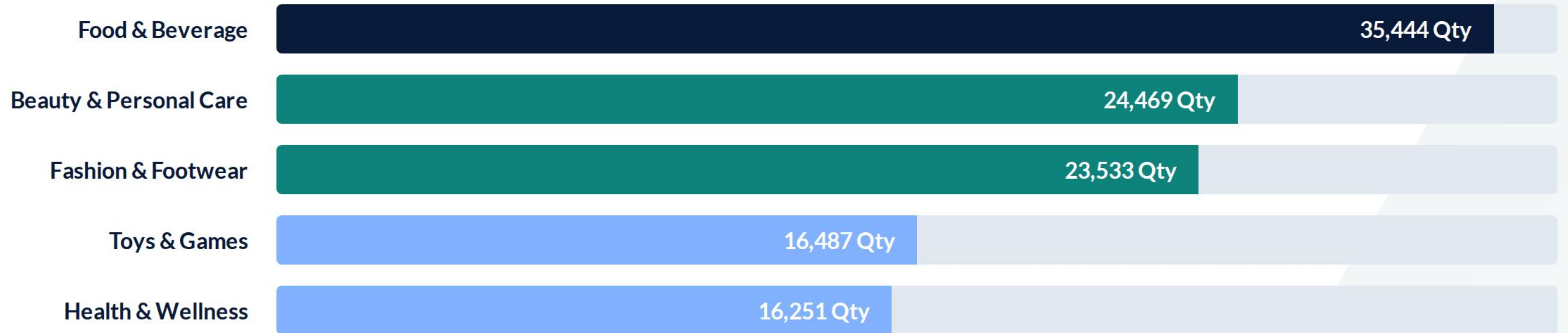
## MONTHLY REVENUE PERFORMANCE (2024)



Revenue reached a peak of Rp 4.12 Billion in December, driven by Year-End campaigns.

# F&B and Beauty Dominate Order Volume

## TOTAL VOLUME PER CATEGORY (2020-2024)

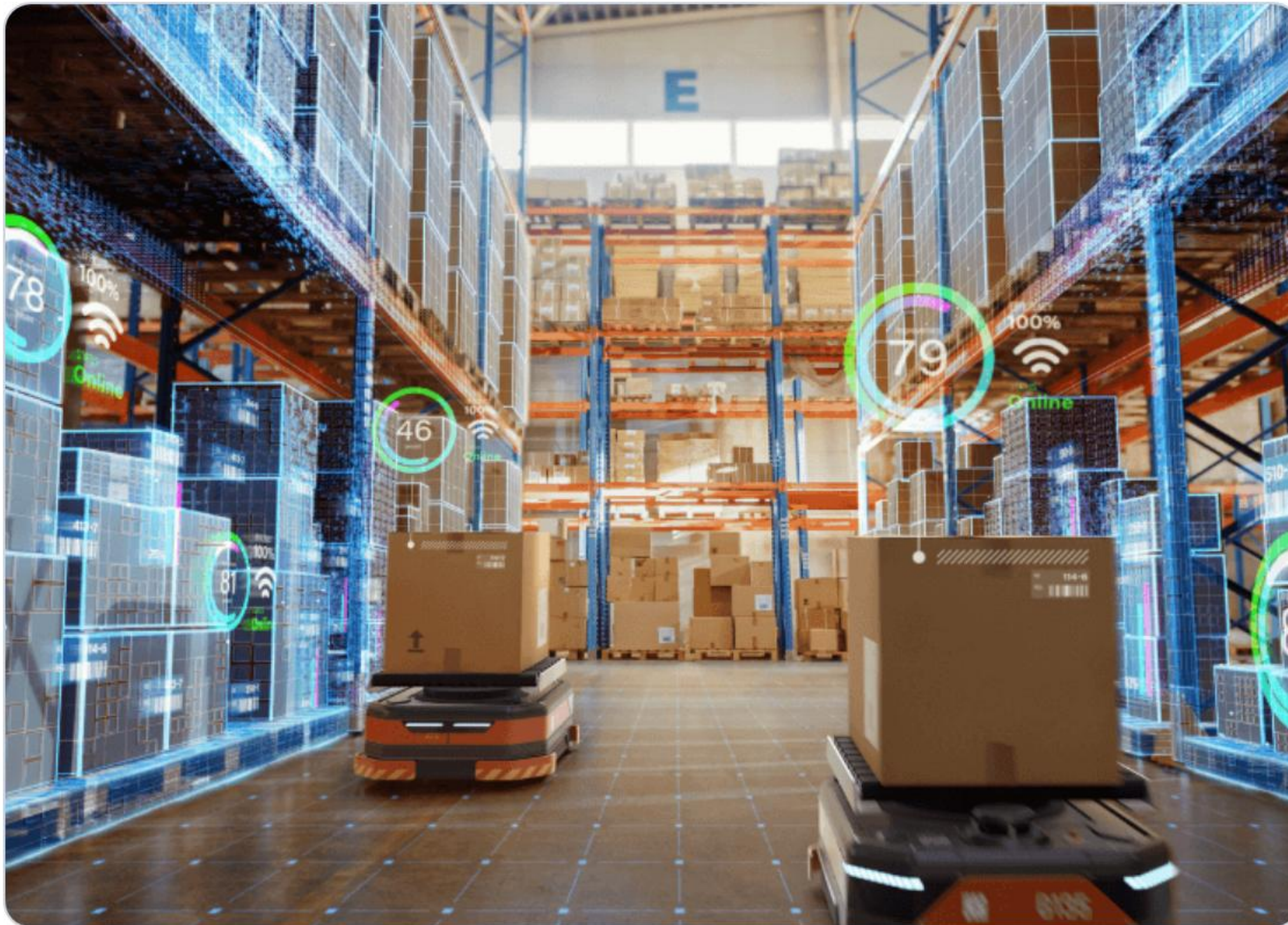


---

# Channel Efficiency & Growth Dynamics

Evaluating the impact of digital vs physical sales channels on annual growth metrics.

# Offline Channels Deliver Highest AOV



## CHANNEL STRATEGIC INSIGHTS

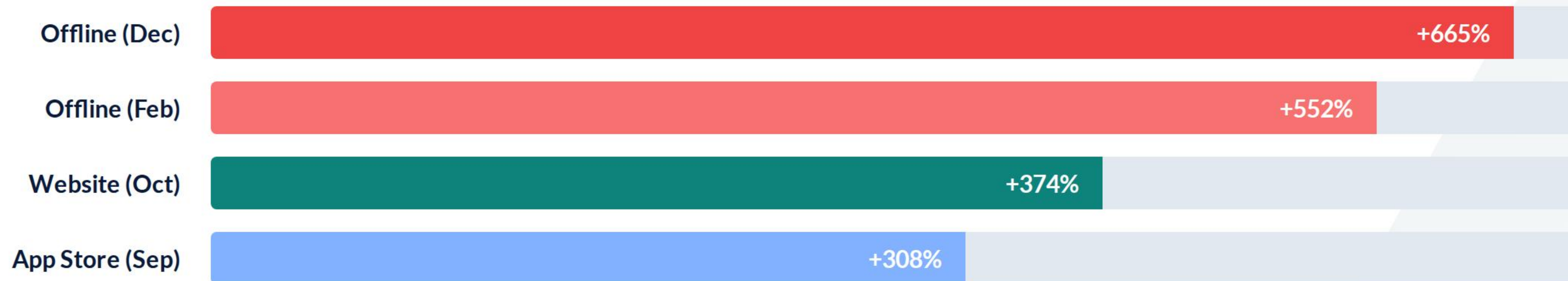
### Revenue by Storefront

While the Mobile App drives consistent volume, **Offline Stores** recorded the highest single-month revenue in December 2024 (Rp 1.48B).

- **Website:** Stable digital anchor.
- **App Store:** Impulsive, promo-driven traffic.
- **Offline:** High-value planned purchases.

# Offline Stores Lead MoM Growth

## REVENUE GROWTH % (2024 VS 2023)



*Exponential growth in Offline channels indicates untapped high-value potential in physical retail presence.*

# Conversion Rate Analysis: Organic Funnel

## ORGANIC TRAFFIC FUNNEL (FY 2024)

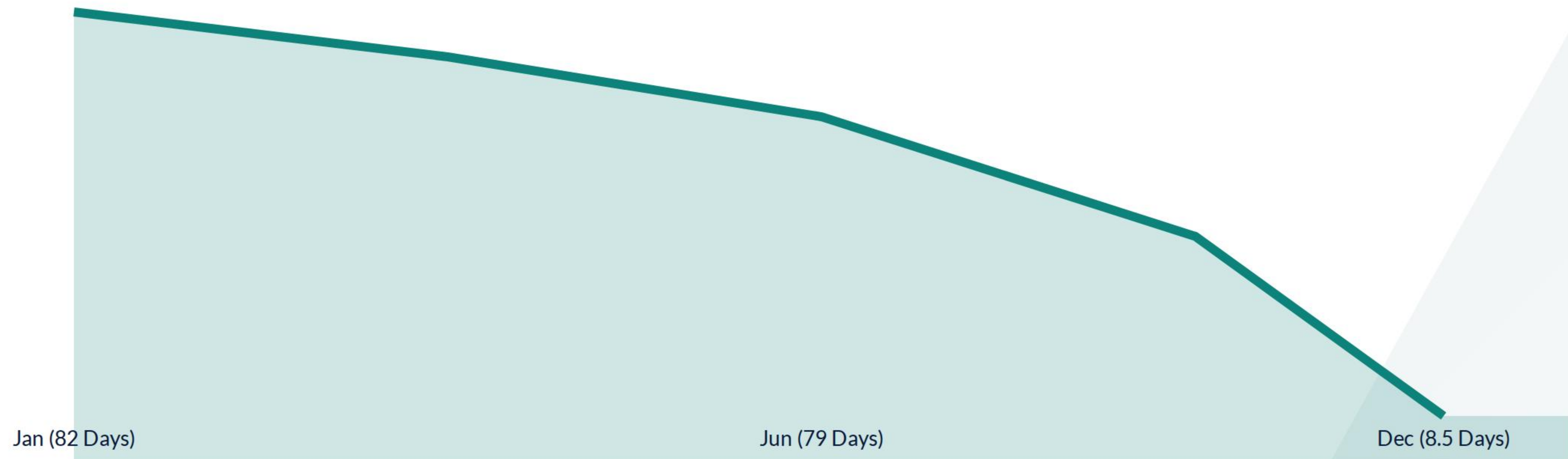


- 2.19% Converted (Purchases)
- 97.81% Awareness (Non-Purchase)

$$CR = \frac{\text{Total Orders}}{\text{Total Events}} \times 100 \%$$

# Onboarding Speed Increased in Q4

AVG. DAYS TO FIRST ORDER (2024 TREND)



**Strategic Shift:** Time-to-First-Order dropped from 82 days in Jan to only 8.5 days in Dec.

# Strategic Recommendations

- ✓ **Optimize App Onboarding:** Implement "First 7-Day" promo triggers to replicate Q4's 8.5-day conversion speed year-round.
- ✓ **Leverage Offline Presence:** Use high Offline AOV as a flagship for high-value product launches and bundling.
- ✓ **Mid-Year Activation:** Introduce "Mid-Year Mega-Sales" in June-July to mitigate the observed mid-year revenue nadir.



# Questions?

---

Prepared by Kadek Ngurah Silayoga

Senior IT Specialist & Management Portfolio

# Image Sources



[https://static.vecteezy.com/system/resources/thumbnails/060/275/693/small\\_2x/a-dynamic-business-meeting-in-a-modern-office-brings-together-diverse-professionals-for-innovation-photo.jpg](https://static.vecteezy.com/system/resources/thumbnails/060/275/693/small_2x/a-dynamic-business-meeting-in-a-modern-office-brings-together-diverse-professionals-for-innovation-photo.jpg)

Source: [www.vecteezy.com](https://www.vecteezy.com)

---



<https://www.lightguidesys.com/wp-content/uploads/2024/06/smart-warehouse-technology.png>

Source: [www.lightguidesys.com](https://www.lightguidesys.com)

---



[https://static.vecteezy.com/system/resources/previews/035/580/970/non\\_2x/growth-bar-charts-financial-graph-on-digital-technology-strategy-background-with-finance-data-marketing-chart-analysis-report-or-success-investment-charts-vector.jpg](https://static.vecteezy.com/system/resources/previews/035/580/970/non_2x/growth-bar-charts-financial-graph-on-digital-technology-strategy-background-with-finance-data-marketing-chart-analysis-report-or-success-investment-charts-vector.jpg)

Source: [www.vecteezy.com](https://www.vecteezy.com)